

PAULS VALLEY, OK, USA

WMO: 722173

Lat: 34.711N

Lon: 97.223W

Elev: 968

StdP: 14.19

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53951

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.7	20.8	3.4	6.8	19.9	9.4	9.2	28.0	26.1	51.2	24.0	49.5	6.7	340	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	100.3	75.1	98.6	75.0	95.2	75.5	79.1	91.8	78.2	91.1	77.4	90.3	8.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.6	138.8	83.7	74.8	135.1	83.3	73.4	128.9	82.4	43.1	91.9	42.2	91.2	41.4	89.6	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	20.3	18.3	DB	9.8	103.7	4.1	3.5	6.9	106.2	4.5	108.2	2.2	110.2	-0.8	112.8
			WB	9.0	81.0	3.8	1.4	6.2	82.0	4.0	82.8	1.9	83.6	-0.9	84.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.0	41.5	45.4	55.1	62.2	70.0	80.0	83.3	82.7	75.7	63.6	52.9	42.8
	DBStd	17.10	10.52	11.63	9.87	8.54	7.40	4.81	4.90	5.78	6.50	9.10	9.50	10.07
	HDD50	935	303	206	60	13	1	0	0	0	0	10	76	266
	HDD65	3004	731	554	329	147	37	0	0	0	5	138	374	689
	CDD50	5686	38	77	218	379	622	899	1031	1013	772	430	164	43
	CDD65	2283	1	5	23	64	193	449	566	548	327	94	12	1
	CDH74	25478	16	44	201	594	1698	4962	7011	6652	3313	867	110	10
	CDH80	12369	1	7	40	141	544	2376	3844	3665	1496	240	15	0
Wind	WSAvg	7.7	8.0	8.8	9.5	9.6	7.9	7.7	6.1	5.7	6.0	7.1	8.1	7.9
Precipitation	PrecAvg	37.0	1.7	1.9	2.8	3.3	5.3	4.4	2.4	4.3	3.8	2.7	2.0	
	PrecMax	58.2	6.7	6.2	10.2	11.2	14.3	12.2	7.0	7.6	11.0	7.6	6.7	
	PrecMin	19.0	0.0	0.0	0.1	0.2	0.2	0.7	0.0	0.0	0.2	0.1	0.1	
	PrecStd	8.3	1.5	1.5	1.8	2.0	3.4	2.7	1.9	2.0	2.7	2.8	2.1	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.4	80.6	84.2	90.1	93.0	99.4	104.8	108.5	99.3	90.3	81.8	75.0
		MCWB	55.8	62.5	65.2	68.2	75.2	74.8	74.8	75.2	72.2	70.9	63.8	60.3
	2%	DB	70.0	73.5	79.4	83.2	89.9	96.7	100.1	101.8	94.8	86.3	77.0	69.5
		MCWB	55.1	59.2	61.4	65.4	74.4	74.8	74.9	75.3	73.4	70.3	62.5	58.4
	5%	DB	63.8	69.7	75.1	80.9	86.3	94.5	98.4	99.0	91.1	82.0	72.5	63.6
		MCWB	52.8	57.5	61.2	64.7	72.2	74.9	75.0	75.6	72.8	67.0	60.4	53.2
	10%	DB	59.3	63.7	71.2	76.9	82.1	90.6	95.0	95.1	89.7	79.1	69.6	59.1
		MCWB	48.0	54.4	58.8	63.4	70.2	75.6	75.9	76.0	72.4	66.1	59.7	50.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.5	66.2	68.2	72.6	77.9	80.3	80.5	79.6	78.4	75.1	69.1	65.7
		MCDB	68.2	72.4	78.3	82.6	89.1	91.2	91.7	92.8	89.5	83.9	76.9	70.6
	2%	WB	58.8	63.5	66.2	70.3	75.6	78.7	79.1	78.5	76.6	73.3	66.1	60.5
		MCDB	65.2	69.5	75.0	78.7	85.2	90.8	91.6	92.6	87.8	81.4	73.0	66.3
	5%	WB	53.5	60.0	64.2	68.4	74.1	77.6	78.1	77.8	75.2	71.2	63.7	56.3
		MCDB	63.6	66.3	71.2	76.5	83.2	89.8	90.8	92.1	85.9	79.3	69.0	60.6
	10%	WB	48.9	55.5	61.7	66.6	72.4	76.5	77.2	76.9	74.1	68.8	60.4	51.5
		MCDB	57.9	64.4	68.8	74.5	80.4	87.8	90.0	90.9	83.5	75.3	67.6	59.1
Mean Daily Temperature Range	5% DB	MDBR	21.6	21.0	21.6	21.8	18.9	19.0	20.2	20.6	21.3	22.5	22.3	19.8
		MCDBR	29.8	28.2	25.8	24.7	21.6	22.5	23.9	24.5	24.4	24.7	26.6	27.6
		MCWBR	17.9	17.1	14.2	13.2	10.3	8.2	6.9	7.1	8.4	11.7	15.6	18.0
	5% WB	MCDBR	26.7	23.5	20.5	20.3	19.1	18.4	19.8	20.3	19.2	19.8	22.2	22.5
		MCWBR	18.6	16.5	13.3	13.5	10.4	8.5	7.7	7.3	7.7	10.6	17.1	18.2
Clear-Sky Solar Irradiance	taub	0.299	0.306	0.346	0.378	0.413	0.425	0.428	0.433	0.389	0.347	0.320	0.298	
	taud	2.503	2.501	2.411	2.341	2.310	2.315	2.343	2.282	2.372	2.483	2.494	2.509	
	Ebn at Noon	290	300	294	288	278	273	271	268	276	281	279	283	
	Edh at Noon	26	30	35	40	41	41	40	42	36	30	26	25	
All-Sky Solar Radiation	RadAvg	886	1066	1406	1729	1851	2108	2146	1956	1620	1241	938	757	
	RadStd	92	161	91	135	195	208	121	115	154	174	119	89	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page